ATTENTION TO INFLUENCE

YouTube Analytics Dashboard Report

How does attention on the internet become influence?

Dataset	78,586 YouTube Trending Videos
Markets	United States · United Kingdom · Canada · India
Tool	Tableau Desktop / Public
Dashboards	4 Dashboards · 16 KPIs · 24 Charts
Total Views Analysed	2.40 Billion

PROJECT TEAM

Bineet Keshari	*bineet.keshari2024@nst.rishihood.edu.in*
Krish Garg	*krish.garg2024@nst.rishihood.edu.in*
MD Sajjan	*md.sajjan2024@nst.rishihood.edu.in*
Mishti Sharma	*mishti.sharma2024@nst.rishihood.edu.in*
Ronit Singh	*ronit.singh2024@nst.rishihood.edu.in*
Srijan Patel	*srijan.patel2024@nst.rishihood.edu.in*

1. Executive Summary

This report documents a four-dashboard Tableau project that traces how internet attention is created, distributed across geographies, sustained through engagement, and ultimately converted into influence. The dataset comprises 78,586 YouTube trending videos across four English-speaking markets: the United States, United Kingdom, Canada, and India.

The central research question is: how does attention on the internet become influence? The project answers this through four sequential dashboards, each examining one stage of the journey from raw visibility to lasting impact.

🔑 **Core Finding**

Attention in the internet era is given before it is earned. Visibility starts the process, but only engagement sustains it — and only deep engagement ever converts into influence. The top 1% of videos (≈786 clips) hold approximately 44% of total views.

Project at a Glance

2.40B	78,586	4	4
Total Views Analysed	Videos in Dataset	Markets Covered	Dashboards Built

2. Dataset & Preparation

The dataset, final_dataset_tableau.csv, contains 78,586 YouTube trending video records spanning four markets. Each row represents a single video's performance snapshot, including raw engagement counts, derived ratios, timing metadata, and content signals.

2.1 Core Dataset Columns

Column	Type	Description
country	String	Market — US, UK, CA, IN
publish_time	Datetime	When the video was uploaded
views	Integer	Total views accumulated
views_log	Float	Natural log of views (for binning)
likes	Integer	Total likes
comments	Integer	Total comments

Column	Type	Description
engagement_rate	Float	(likes + comments) / views
like_ratio	Float	Likes / views
comment_ratio	Float	Comments / views
virality_score	Float	Composite score (0–1)
title_length	Integer	Characters in title
title_uppercase_ratio	Float	Share of uppercase characters
tag_count	Integer	Number of tags on the video
publish_hour	Integer (0–23)	Hour of upload
publish_day	String	Weekday name
is_weekend	Binary	1 if Saturday or Sunday
time_to_trend	Integer	Days from upload to trending

2.2 Key Calculated Fields

The following calculated fields were created in Tableau to power the dashboards:

- Day Name — DATENAME('weekday', [publish_time])
- Weekend Label — IF [is_weekend]=1 THEN "Weekend" ELSE "Weekday" END
- Tag Bucket — grouped into: 0, 1–5, 6–15, 16–30, 31+
- Top 1% Flag — IF virality_score >= PERCENTILE(virality_score, 0.99) THEN "Top 1%" ELSE "Rest" END
- Virality Tier — Viral / Strong / Mid / Low by percentile bands

3. Key Insight Foundation

Before designing the dashboards, exploratory data analysis produced seven anchor findings that drive every chart choice in the project. These findings define the analytical spine of the entire report.

#	Question	Answer	Evidence
1	How does attention start?	Visibility, not quality	views ↔ virality r = 0.73–0.86
2	Do people engage with all attention?	No — most attention is shallow	virality ↔ engagement r ≈ 0.01
3	What separates shallow from real?	Engagement ratios	engagement ↔ like_ratio r = 0.99
4	First signal content is working?	Likes	likes ↔ views r = 0.80

#	Question	Answer	Evidence
5	What shows deeper connection?	Comments	comments ↔ comment_ratio $r = 0.57$
6	What boosts or hurts attention?	Hour, day, title, tags	weekend $t = -27.9$, $p \approx 0$
7	Can we fully predict virality?	Only partially	Model $R^2 = 0.56$

4. Dashboard 1 — What Creates Attention

Purpose: Identify the structural and behavioural levers that allow a video to accumulate raw views. This is the 'spark' stage — exposure before evaluation. The unit of analysis is impressions, and the question is: what shifts the baseline odds of being seen?

4.1 KPIs

2.40B	30.5K	18:00	−27.9
Total Views Analysed	Avg Views / Video	Peak Publish Hour	Weekend t-statistic
Denominates all ratios	*Median: 5.2K (skewed)*	*6 PM local time*	*Significant penalty*

4.2 Charts & Insights

Chart 1 · Views Distribution (Histogram — Log Bins)

Log binning is the only honest way to display a winner-takes-all distribution. Linear binning would compress 99% of videos into the first bar. The resulting histogram shows a strong right skew — the Pareto shape — where most videos cluster at low views while a thin long tail produces virality.

Chart 2 · Weekend vs Weekday (Donut)

71% of posts happen on weekdays, and those weekday posts capture a disproportionate share of total views. The weekend penalty is confirmed by the t-statistic of −27.9, which is statistically decisive. Weekend timing carries a measurable, systematic disadvantage.

Chart 3 · Avg Views by Publish Hour (Area Line)

Two peaks emerge: a late-morning local peak and an evening peak at 18:00. Pre-dawn hours (3–6 AM) collapse to near zero. The area-line encoding conveys both the cyclic shape and the magnitude, confirming the evening window as the dominant upload slot.

Chart 4 · Views by Day of Week (Vertical Bar)

Friday is the highest-performing day. Saturday and Sunday drop sharply, mirroring the weekend t-statistic. The bars allow fast visual ranking of all seven days simultaneously.

Chart 5 · Title Length vs Views (Density Scatter)

A polynomial trend line over a density scatter of 78,000 points reveals a non-linear sweet spot. Titles between 40–55 characters perform best. Too short reads vague; too long reads cluttered and may be penalised by the recommendation algorithm.

Chart 6 · Tag Count Buckets — Avg Views (Horizontal Bar)

Views rise with tag count up to the 16–30 bucket, then flatten and slightly fall. Over-tagging (31+) signals spam to the algorithm and begins to erode discovery. The optimal zone is 16–30 tags.

🔑 Dashboard 1 Takeaway

Attention is given before it is earned. Distribution is Pareto-like, not Gaussian. Timing shifts the baseline odds meaningfully — 6 PM beats 4 AM by multiples, and weekends carry a measurable penalty. Packaging also matters: titles have a 40–55 character sweet spot, and tags show diminishing returns after 30. None of these levers force virality, but all of them move the starting odds.

5. Dashboard 2 — Attention Across Countries

Purpose: Compare how attention behaves across the four markets. A single global strategy misses the divergence between volume leadership and engagement leadership — which do not align across geographies.

5.1 KPIs

4	India	UK	78.5K
Countries Analysed	Top by Total Views *≈38% of total views*	Top by Engagement Rate *≈4.8% avg engagement*	Total Videos in Panel

5.2 Charts & Insights

Chart 1 · World Choropleth

Geographic fill by SUM(views) on a sequential red ramp makes the volume hierarchy instantly legible. India fills darkest; the US fills bright red; UK and Canada appear mid-tone. No table or label is needed to read the ranking.

Chart 2 · Virality Score by Country (Vertical Bar)

India 0.91 → US 0.74 → UK 0.58 → Canada 0.47. The virality ordering follows the volume ordering — high-reach markets also produce more viral outliers, likely because larger audiences create stronger recommendation loops.

Chart 3 · Engagement Rate by Country (Lollipop)

UK 4.8% > India 4.1% > US 3.2% > Canada 2.8%. This ranking inverts the volume ordering. The UK — which ranks third in total views — leads on engagement rate. Volume leadership and engagement leadership are decoupled.

Chart 4 · Share of Total Views (Stacked Donut)

India 38%, US 29%, UK 18%, CA 15%. Two-country dominance is visible at a glance, with the donut serving as a visual cross-reference to the KPI values.

Chart 5 · Publish Hour × Country Heatmap

The 18:00 peak is universal, but the cycle shifts east-to-west with time zone. India peaks earliest in UTC; the US peaks latest. A global upload strategy that uses a single absolute time will miss the peak window in multiple markets simultaneously.

Chart 6 · Weekend Share by Country (100% Stacked Bar)

India creators post most on weekdays. UK and Canada post a higher proportion of content on weekends. Because the comparison is normalised to 100%, it reflects behavioural differences rather than sample-size differences.

🔑 Dashboard 2 Takeaway

Geography reshapes both the scale and texture of attention. Volume leaders are not engagement leaders. The evening peak is universal, but its UTC timing shifts east-to-west. Weekend behaviour varies enough across countries that a global 'avoid weekends' rule is too blunt — market context is required for the strategy to be actionable.

6. Dashboard 3 — What Sustains Attention

Purpose: Once a video is seen, does it provoke a response? This dashboard switches the unit of analysis from impressions to reactions. Most attention is shallow, and only a small fraction of views convert into likes — and an even smaller fraction into comments.

6.1 KPIs

3.7%	3.4%	0.32%	0.041
Avg Engagement Rate	Avg Like Ratio *Dominates engagement*	Avg Comment Ratio *10× rarer than likes*	Retention Index (median)

6.2 Charts & Insights

Chart 1 · Engagement Rate Distribution (Histogram)

The distribution concentrates below 5% with a long right tail that mirrors the views histogram. A median reference line at 3.7% sits well left of the mean, confirming right skew. The same Pareto shape that governs views governs engagement.

Chart 2 · Views vs Likes (Log–Log Scatter)

Correlation r = 0.80. More views reliably produce more likes across all orders of magnitude. This confirms the algorithmic feedback loop: exposure generates reactions, which in turn boost recommendation priority.

Chart 3 · Views vs Comments (Log–Log Scatter)

Correlation r = 0.67. Comments scale with views but with more noise and a shallower slope. Depth does not keep up with reach — the comment signal is genuinely rarer and therefore carries more per-unit informational value.

Chart 4 · Engagement by Publish Hour (Heatmap)

Engagement peaks midweek at 18:00. The concentration of engagement is tighter than the spread of reach — meaning the optimal engagement window is more specific than the optimal reach window. Both agree on evenings; engagement is more punishing of off-peak timing.

Chart 5 · Like Ratio vs Comment Ratio (Bubble)

The two ratios move together at low values but decouple at the top. Viral videos split into two archetypes: 'clicky and entertaining' (high likes, average comments) and 'discussed and divisive' (high comments, moderate likes). These represent two distinct content playbooks.

Chart 6 · Tag Count → Engagement Lift (Combo Bar + Line)

Both reach and engagement peak at 15–25 tags. Beyond that, views continue rising slightly but engagement falls. Over-tagging signals spam to the algorithm and dilutes the conversation that would otherwise form around focused content.

🔑 Dashboard 3 Takeaway

Most attention is shallow. Engagement concentrates below 5%, and comments are an order of magnitude rarer than likes. The strong views–likes correlation confirms likes are mostly a scale signal. The weaker views–comments correlation tells us depth does not scale automatically. At the top, 'clicky' and 'discussed' videos diverge into two different playbooks.

7. Dashboard 4 — Attention into Influence

Purpose: Close the loop. Influence is the rarest outcome, and the unit of interest shifts from average behaviour to the tail of the distribution. The central question is: what separates the videos that merely trend from those that become influential?

7.1 KPIs

0.62	786	1.8 days	0.56
Avg Virality Score	Top 1% Videos	Avg Time to Trend	Model R^2 (Predictability)
Heavy tail above mean	*Hold ~44% of total views*	*Early or never*	*Half is learnable*

7.2 Charts & Insights

Chart 1 · Attention → Influence Funnel

A centred funnel visualises the steep drop-off at each conversion stage: 2.40 B views → 88.6 M engaged views → 8.1 M deeply engaged views → 786 influential videos. Each stage drops by 10–100×. The cliff from deep engagement to influence is the steepest transition.

Chart 2 · Virality Score Distribution (Histogram)

The body of the distribution is left-heavy; the top 1% bar is coloured amber and sits far to the right. The mean of 0.62 lies between the two concentrations, illustrating why mean virality is a misleading summary statistic for this dataset.

Chart 3 · Top Drivers of Virality (Coefficient Bars)

The ranked coefficient chart provides a direct decision list for creators. Likes (+0.80) and views (+0.73) dominate as positive drivers. Weekend posting (−0.19) and uppercase-heavy titles (−0.12) are the most significant negative signals. Tag count is mildly positive (+0.28). Two-colour encoding (red positive, dark negative) makes direction immediately legible.

Chart 4 · Time to Trend Distribution (Histogram)

Most trending happens inside 2 days of upload. After one week, odds collapse toward zero. The distribution is highly right-skewed with a hard practical cutoff around day 7. Attention is impatient — videos either gain momentum immediately or miss the window entirely.

Chart 5 · Influence vs Views (Tier-Coloured Scatter)

Four virality tiers (Viral / Strong / Mid / Low) are colour-coded on a log–log scatter of influence against views. The Viral tier clusters tightly in the top-right quadrant. The Low tier fills the floor. Tier separation is clean enough to be meaningful for targeting and filtering decisions.

Chart 6 · Top Viral Videos Treemap

Tile area and colour intensity both encode virality_score, so the most influential content is both largest and most saturated. A handful of Indian and US videos dominate the area. The influence set is geographically concentrated and dramatically outscales the next tier.

🔑 **Dashboard 4 Takeaway**

Influence is a rare conversion. The top 1% of videos (~786 clips) carry approximately 44% of total views. The feature importance bar gives creators a short decision list: optimise for like-generating hooks, avoid weekend drops, avoid uppercase clickbait titles. The R^2 of 0.56 is the honest caveat — better choices roughly double the odds, but no model closes the gap between strong content and true virality.

8. Consolidated Narrative

Read end to end, the four dashboards answer one question: how does an ordinary video become an influential one? The data splits the journey into four stages, each with its own drop-off rate and distinct design implications for creators and platforms.

Stage	Dashboard	Drop-off	What the Creator Controls
1. Visibility	D1 — What Creates Attention	Entry point	Hour, day, title length, tag count
2. Geographic Reach	D2 — Attention Across Countries	~Uniform per market	Local timing, weekend strategy
3. Engagement Depth	D3 — What Sustains Attention	Views → reactions ≈ 4%	Hooks, format, tag balance
4. Influence Conversion	D4 — Attention into Influence	Engagement → influence ≈ 1%	Speed (≤2 days) + structural levers

Stage-by-Stage Summary

Stage 1 — Visibility

Distribution is Pareto. The mean is ~6× the median, confirming winner-takes-all dynamics. Timing and packaging shift baseline odds but cannot force virality. This is the most controllable stage: post on a weekday, aim for 6 PM local, keep titles 40–55 characters, and use 16–30 tags.

Stage 2 — Geographic Reach

Volume leadership does not equal engagement leadership. India leads on raw views and virality score; the UK leads on engagement rate. Time-zone shifts alter the optimal absolute upload hour by market. A single global rule is insufficient — market-aware strategy is required.

Stage 3 — Engagement Depth

Most attention is shallow: median engagement rate ≈ 4%. Likes scale strongly with views ($r = 0.80$); comments scale more weakly ($r = 0.67$). At the top of the distribution, content splits into two archetypes — 'clicky/entertaining' and 'discussed/divisive' — requiring different production approaches.

Stage 4 — Influence Conversion

Only ~786 videos (the top 1%) cross into the true viral tier, holding ~44% of all views. Time-to-trend is decisive: winners emerge inside 1–2 days. The model explains 56% of virality variance — roughly half is structural and learnable; the other half is emergent and unpredictable.

9. Design System

All four dashboards use a consistent YouTube dark theme to ensure visual coherence across the project.

9.1 Colour Tokens

Token	Hex Code	Usage
Canvas Background	#0F0F0F	Entire dashboard background
Card Background	#1C1C1C	KPI cards, chart containers
Primary Accent Red	#FF0000	Main bars, primary highlights
Red Shade 1	#FF3333	Histogram bars
Red Shade 2	#E53935	Secondary bars
Red Shade 3	#B71C1C	Heatmaps
Amber Highlight	#FFD54F	Outliers / Top 1% viral callouts
Positive Green	#4CAF50	Positive KPI deltas
Primary Text	#FFFFFF	Titles, KPI values
Secondary Text	#AAAAAA	Axis labels, subtitles

9.2 Layout Grid

- Fixed dashboard size: 900 × 990 px
- Header row: 72 px tall
- KPI row: ~104 px tall, 4 cards across
- Chart rows: 3 rows of ~250–270 px each, 1–2 charts per row

9.3 Interaction Design

- Country filter active on every dashboard
- Time-range filter active on every dashboard
- Cross-filter actions: clicking a country bar or tile filters the rest of the dashboard
- Highlight actions where filtering would empty the view

10. Limitations & Next Steps

10.1 Current Limitations

- Single platform: findings generalise to YouTube only. Replication on TikTok or Instagram would test whether the Pareto shape is platform-specific or universal.
- Four English-speaking markets: behaviour from non-English-speaking regions may differ significantly. Expanding the geographic panel is the highest-value near-term extension.
- Snapshot, not survival: time-to-trend is treated as a static field. A true survival model with hazard rates would likely lift the R^2 above 0.56.

- No content category dimension: a topic/category column would allow the engagement split to be read by content vertical, which is the most-requested analytical cut from creators.
- No causal inference: all coefficients are correlational. An A/B test on publish-hour or title-length would convert the findings from explanatory to prescriptive.

10.2 Recommended Next Steps

- Extend to 8–10 markets including non-English-speaking regions (Brazil, Japan, Germany).
- Add a content category dimension to the dataset to enable vertical-level analysis.
- Build a survival model for time-to-trend using hazard rates rather than static averages.
- Run controlled A/B experiments on publish-hour and title-length to establish causal direction.
- Replicate the pipeline on TikTok data to test cross-platform generalisability.

Appendix — KPI & Chart Inventory

Complete reference of all 16 KPIs and 24 charts across the four dashboards.

Dashboard	KPIs	Chart 1	Chart 2	Chart 3	Chart 4	Chart 5	Chart 6
D1 — Creates	Total views, Avg views, Peak hour, Weekend t-stat	Views histogram	Weekend donut	Hourly area line	Day-of-week bars	Title-length scatter	Tag-bucket bars
D2 — Countries	Countries, Top by views, Top by engagement, Total videos	World choropleth	Virality bars	Engagement lollipop	Share donut	Hour×country heatmap	Weekend share stacked
D3 — Sustains	Avg engagement, Avg like ratio, Avg comment ratio, Retention	Engagement histogram	Views vs likes scatter	Views vs comments scatter	Engagement heatmap	Like vs comment bubble	Tag count combo
D4 — Influence	Avg virality, Top 1% count, Time to trend, Model R^2	Conversion funnel	Virality histogram	Feature importance bars	Time-to-trend histogram	Influence vs views scatter	Viral treemap

— End of Report —